Suvinava Basak

Braunschweig, Germany | suvinava.basak@outlook.com | suvinava.basak@tu-braunschweig.de | Homepage

Research Interests

Computational systems biology and systems medicine, with a focus on machine learning for bio-data integration, biological network modelling, and personalized computational modelling. My research interests include integrating heterogeneous molecular and genomic data, including transcriptomic and multi-omics information, with biological knowledge and pathway representations to investigate complex disease mechanisms and identify predictive biomarkers and therapeutic targets.

Education

Technische Universität Braunschweig, Germany *Oct 2023 – Ongoing*
Master of Science (M.Sc.) in Data Science [Transcript] [Relevant Coursework]
Thesis: *End-to-End Calibration: Theory and Empirical Evaluation in Biomedical Context* [Thesis Report] [Repository]
Thesis advisor: Prof. Dr. Nicole Mücke | **Current grade: 1.3 (German scale, 1.0 = best)**, excl. thesis. Exp. Grad: Nov 2026

Indian Institute of Management Indore, India *Jul 2016 – Mar 2018*
Master of Business Administration (MBA) [Transcript] [Degree]

Jalpaiguri Govt. Engineering College, West Bengal University of Technology, India *Aug 2010 – Jun 2014*
Bachelor of Technology in Computer Science and Engineering [Transcript] [Degree] [Coursework]
Project: *Face Recognition System using MATLAB* [Report] [Repository]
Advisor: Prof. Animesh Hazra | **Grade: 7.83/10.0** | **Top 10%** (of 60 students)

Research Experience

Institute for Statistics and Foundations of Machine Learning, TU Braunschweig *Apr 2026 – Ongoing*
Master's Thesis | Advisor: Prof. Dr. Nicole Mücke
End-to-End Calibration: Theory and Empirical Evaluation in Biomedical Context

- Investigating end-to-end calibration of neural networks, incorporating differentiable calibration objectives directly into model training rather than relying only on post-hoc calibration, and studying their effects on predictive reliability and uncertainty estimation.
- Studying the relationship between predictive performance, model confidence and reliability under distribution shift, using Transformer-based gene regulatory network inference as a scientific application.

CAIMed Research Group, Medizinische Hochschule Hannover, Germany *Aug 2025 – Ongoing*
Student Research Assistant | Advisor: Dr. rer. nat. Zahra Ahmadi, Dr.-Ing. Raneen Younis [Reference Letter]
Multi-agent System for Scientific Hypothesis Generation

- Developing an LLM- and Knowledge Graph-aware RAG-based multi-agent system for drug combination hypothesis generation from heterogeneous biomedical evidence, integrating molecular, genomic, clinical and literature information.
- Designing evidence-grounded reasoning and calibrated confidence mechanisms for reliable scientific decision-making reasoning under incomplete or heterogeneous evidence.

Institute for Statistics and Foundations of Machine Learning, TU Braunschweig *Mar 2026 – May 2026*
Student Research Assistant | Advisor: Prof. Dr. Nicole Mücke
Calibrating Neural Network under Distribution Shift

- Researched calibration and uncertainty estimation of deep networks and its performance under distribution shift.

Publications

Accepted Papers

- **S. Basak**, *"Is grokking a loss of normal hyperbolicity of the interpolation manifold?"*, 2026. Oral presentation at SKILL 2026 (Studierendenkonferenz Informatik), INFORMATIK 2026, Gesellschaft für Informatik e.V., Dresden, 23 September. To appear in GI-Edition Lecture Notes in Informatics (LNI). [Paper] [Presentation]
- R. Younis*, **S. Basak***, L. Chavez, and Z. Ahmadi, *"From literature to hypotheses: An AI co-scientist system for biomarker-guided drug combination hypothesis generation"*, 2026. Accepted at IEEE ICDM 2026, Demo track. To appear in ICDM 2026 Workshop Proceedings. [Paper] [Demo]

Teaching Experience

Institute for Analysis and Algebra, TU Braunschweig *Apr 2025 – Sep 2025*
Student Research Assistant (Teaching Assistant) | Lecturer: Prof. Dr. Timo de Wolff

- Led tutorials and graded assignments for Mathematical Algorithms and Programming 2 (bachelor level course) in summer semester 2025 for ~50 students. [Reference Letter]

Academic Research Projects

Uncertainty Quantification in Neural Networks *(Gaussian Processes, Distribution Shift)*
Investigated uncertainty estimation in neural networks through the connection between finite-width neural networks and infinite-width Gaussian-process limits. Compared NNGPs, deep ensembles, MC dropout, and RBF Gaussian processes, examining their predictive uncertainty and behavior under distribution shift. [GitHub]

Grokking and Normal Hyperbolicity of Interpolation Manifold *(Grokking, PyTorch, Neural Networks)*
Investigated possible interpretation of sharp generalization transition in grokking as a loss of normal hyperbolicity of the interpolation manifold. Experimental evaluation of the hypothesis through a singular-value-based diagnostic of residual Jacobian. Accepted as a paper at SKILL2026 student conference by German Informatics Society. [GitHub]

Continuous Optimization DAG Learning *(Causal Discovery, DAG, Statistical learning)*
Audited continuous-optimization approaches to Directed Acyclic Graphs learning, including NOTEARS/DAGMA, with emphasis on benchmark sensitivity, standardization and causal-structure recovery. Compared learned graph structures and explored a directed-to-undirected bridge on real protein-signaling data. [GitHub]

Interpretable Cancer Drug-Response Prediction *(Precision Oncology, GNN, Multi-omics)*
Developed an interpretable deep-learning pipeline for cancer drug-response prediction, integrating Gene Ontology-guided neural networks with molecular graph-based drug representations. Implemented drug-aware gene gating, multimodal fusion, and pathway-level interpretability, with synthetic-data experiments and a GDSC2 data pipeline. [GitHub]

Covariate Shift Regularization using Kernel Methods *(Gradient Descent, Distribution Shift)*
Investigated how covariate shift affects model generalization and predictive reliability under distribution mismatch; evaluated importance-weighting and self-attention-based approaches for adapting to changing input distributions. Student project supervised by Prof. Dr. Nicole Mücke, **graded 1.0**. [GitHub]

Binary AMR Prediction from pbp4 Gene using Transformers *(DNABERT, Nucleotide Transformer, PEFT)*
Benchmarked encoder-only pretrained transformer models for binary antimicrobial resistance prediction in *Staphylococcus aureus* against *Cefoxitin* using *pbp4* gene sequences. Analyzed learned sequence representations and classification performance across architectures, with LoRA-based parameter-efficient fine-tuning to adapt pretrained biological sequence models. Seminar "*Deep Learning in Molecular Biology*", supervised by Prof. Dr. Alice C. McHardy. [GitHub]

Industry Experience

Intern, Data Analytics & AI, Volkswagen Financial Services AG, Germany	*Aug 2024 – Feb 2025*	[Exp. Letter]
Strategy Consultant, Wipro Ltd., India	*Jun 2018 – Oct 2023*	[Exp. Letter]
Solution Integrator, Ericsson Global India Services Pvt. Ltd., India	*May 2015 – Jun 2016*	[Exp. Letter]

Technical Skills

Programming	*Python • Julia • SQL • MATLAB*
Deep Learning	*Neural networks • Transformers • Representation learning • Parameter-efficient Fine-tuning (PEFT)*
Statistical & Probabilistic ML	*Uncertainty quantification • Calibration • Gaussian processes • Statistical learning • Distribution Shift • Causal discovery*
Scientific ML	*Causal discover • Graph learning • Multi-modal learning*
Libraries	*PyTorch • TensorFlow • Hugging Face • NumPy • Pandas • Scikit-learn*
LLM Systems	*LLM APIs • RAG • Multi-Agent Systems • Knowledge Graphs • LangChain*
Tools	*Git • Docker • Linux • LaTeX*
Spoken Languages	*English (Proficient) • Bengali (Native) • Hindi (Proficient) • German (Beginner)*

Awards & Recognition

- **Deutschlandstipendium (2025/26)** — merit-based scholarship for high-achieving and committed students.
- Industry recognition, Wipro Ltd. — Excellence Award (Q3–Q4, FY19); Customer Centricity Award (Q1, FY23).

References

Prof. Dr. Nicole Mücke
Technische Universität Braunschweig
nicole.muecke@tu-braunschweig.de
Master's Thesis Supervisor

Dr. rer. nat. Zahra Ahmadi
Medizinische Hochschule Hannover
ahmadi.zahra@mh-hannover.de
Research Group Leader

Dr.–Ing. Raneen Younis
Medizinische Hochschule Hannover
younis.raneen@mh-hannover.de
Research Supervisor